INSURANCE PRODUCT PREDICTION MODELS

TEAM 13
LIAM CAWLEY, SEAN FULCHER, GRAY NICO, SASHA VOLODIN

25th November 2025

Table of Contents

Overview.....1

Methodology.....1

Data Preparation.....1

Analysis.....1

Neural Network.....1

Model Comparison.....2

Results & Recommendations.....4

Conclusion.....4

Appendix.....5

INSURANCE PRODUCT PREDICTION MODELS

Overview

The Department of Customer Services and New Products at the Commercial Banking Corporation (hereinafter referred to as the "Bank") aims to utilize a powerful predictive model to identify which customers are likely to purchase a variable rate annuity product. To ensure that the Bank is provided with an optimal model that retains some interpretability while prioritizing accuracy, our team used a variety of predictive approaches. **We recommend that the Bank use the random forest (RF) model, which correctly distinguishes between who will and will not buy the product nearly 80% of the time.**

Methodology

This section outlines our data preparation for modeling.

Data Preparation

For this phase of the project, our team utilized a training dataset with 8,495 observations and a validation dataset with 2,124 observations. Each row represents an existing customer who was offered the annuity product. **The target variable indicates whether the customer purchased the insurance product or not.** Customer attributes were organized into 37 additional variables, and any numeric variable with ten or fewer distinct values was treated as categorical, excluding the branch variable. All missing variable values were imputed using the **median of the training dataset**, and **15 additional flags were added** as variables to indicate when a value was imputed in their corresponding column. Preliminary variable selection was not necessary, as the Bank already identified the appropriate candidate predictors.

Analysis

This section describes the development of our neural network model, as well as our comparisons of all models developed for the Bank.

Neural Network

To provide the Bank with an option that can properly handle complex relationships between predictors, our team developed a neural network (NN) to enhance prediction accuracy. We created a tuning grid using cross-validation to determine the optimal regularization strength and the number of hidden layers, thereby preventing overfitting. Based on the grid's results, we used **five hidden layers and a regularization strength of 1.**

Finally, we calculated a ROC curve (Figure A1) using the validation data and classified any customer with **a purchase probability of at least 0.36** as a potential buyer of the variable

annuity product. After using this cutoff for predictions, we found that the NN correctly distinguished between those who did or did not buy the product **78.46% of the time.**

Model Comparison

Throughout the three phases of this project, our team has developed **six different models** to explore their respective advantages and disadvantages. Table 1 summarizes the performance metrics of each model after it was tested on validation data, ranked by descending AUC.

Table 1: Summary of all models' performance metrics.

Model	AUC	True Positive Rate	False Positive Rate	Specificity
XGBoost (XGB)	79.68%	79.65%	33.86%	66.14%
Random Forest (RF)	**79.49%**	**75.74%**	**28.87%**	**71.13%**
Neural Network (NN)	78.46%	74.66%	28.15%	71.85%
GAM	78.31%	68.73%	23.95%	76.05%
Logistic Regression (LR)	78.20%	77.22%	31.91%	68.09%
MARS	78.18%	78.57%	33.14%	66.86%

Based on Table 1, no model was analytically the best or worst across all four metrics. We thus calculated the average rank of each model based on all performance metrics. **The RF had an average rank of 2.75, which was the best rank overall.** It also strikes **a good balance** between a high true positive rate (TPR) and a high specificity (i.e., the true negative rate). Other models, like the GAM or MARS, leaned too heavily towards only one of these metrics. Therefore, **our team selected the RF model as our final choice due to its overall performance.**

Figure 1 visualizes the mean decrease accuracy (MDA) of the top ten predictors in the RF.

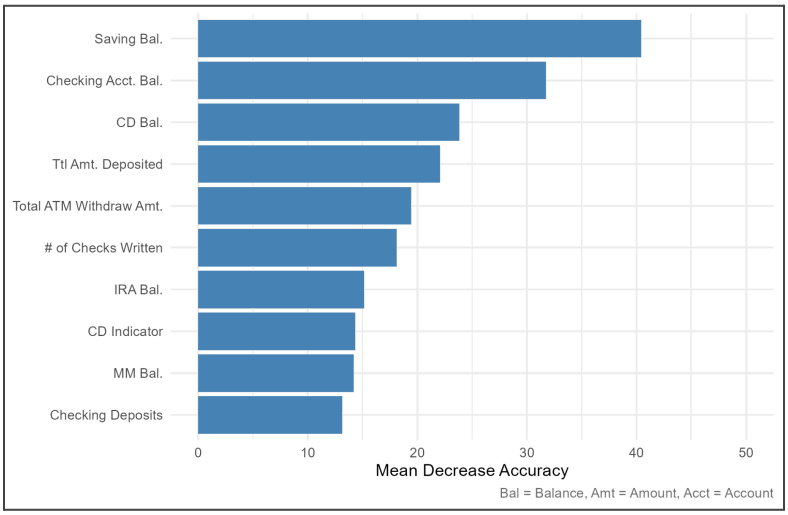

Figure 1: Importance of final RF model predictors based on mean decrease accuracy

From Figure 1, we see that **a customer's savings balance, checking account balance, CD balance, total amount deposited, and total amount of ATM withdrawals** are the strongest predictors of whether that customer will buy the annuity product or not.

Figure 2 visualizes the RF's ROC curve on the validation set, which produced the performance metrics described in Table 1 for this model.

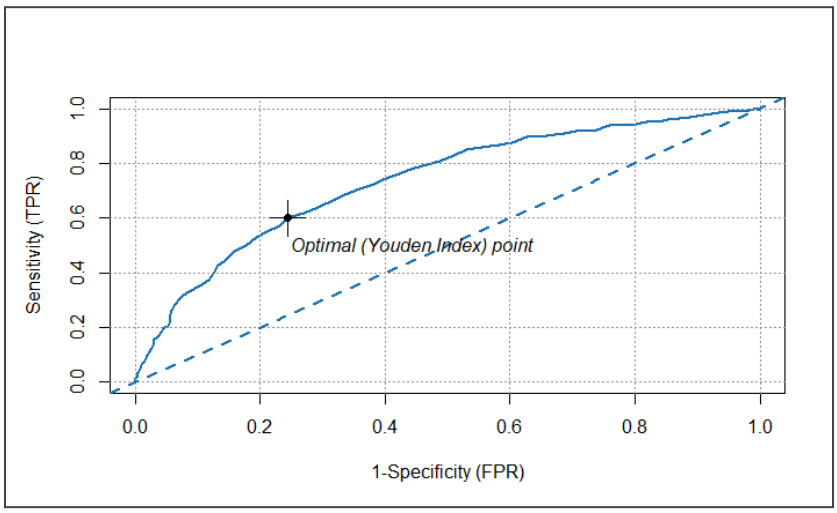

Figure 2: ROC curve for final random forest predictions on validation data

Based on the AUC value in Table 1 and the ROC curve in Figure 2, the RF correctly distinguished between out-of-sample customers who did or did not buy the product **79.49% of the time.**

While account age was not a significant contributor to the RF's overall accuracy, the Bank expressed an interest in measuring its value. Figure 3 visualizes the partial dependence of the random forest predictions on account age, **enabling us to identify how predictions vary with a customer's account age (in years).**

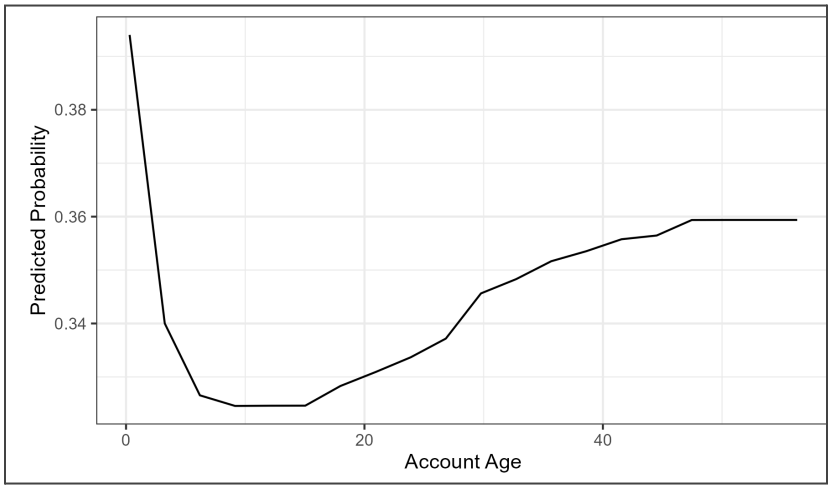

Figure 3: Partial dependence plot of purchase predictions on customer account age

Based on Figure 3, brand new accounts are predicted to have the highest product purchase probability. However, purchase probability declines as accounts age up to ten years. After ten years, the probability increases almost linearly before plateauing around 50 years. This result indicates that **the Bank should target customers with either a very new account or an account that is over 45 years old.** It is essential to note that predicted probabilities do not vary significantly with account age (within an approximate range of 0.3 to 0.4), indicating that **account age has a relatively weak effect on the predicted probability.**

Results & Recommendations

Of the six models our team has developed for predicting whether a customer will purchase the insurance product, we recommend that the Bank utilize the random forest model. **With the RF model, which is both conservative and accurate, the Bank will neither over-market to non-buyers nor ignore too many potential buyers.**

Conclusion

We believe that every false positive represents a short-term loss for the Bank, as they would spend money marketing to a non-buyer. However, every false negative represents a long-term loss for the Bank, as it would fail to acquire a buyer that could have generated more revenue over time. By utilizing our balanced random forest model, **the Bank will be able to accurately identify and invest in customers who are more likely to purchase the variable rate annuity product**, thereby helping the Bank reduce losses and maximize returns.

Appendix

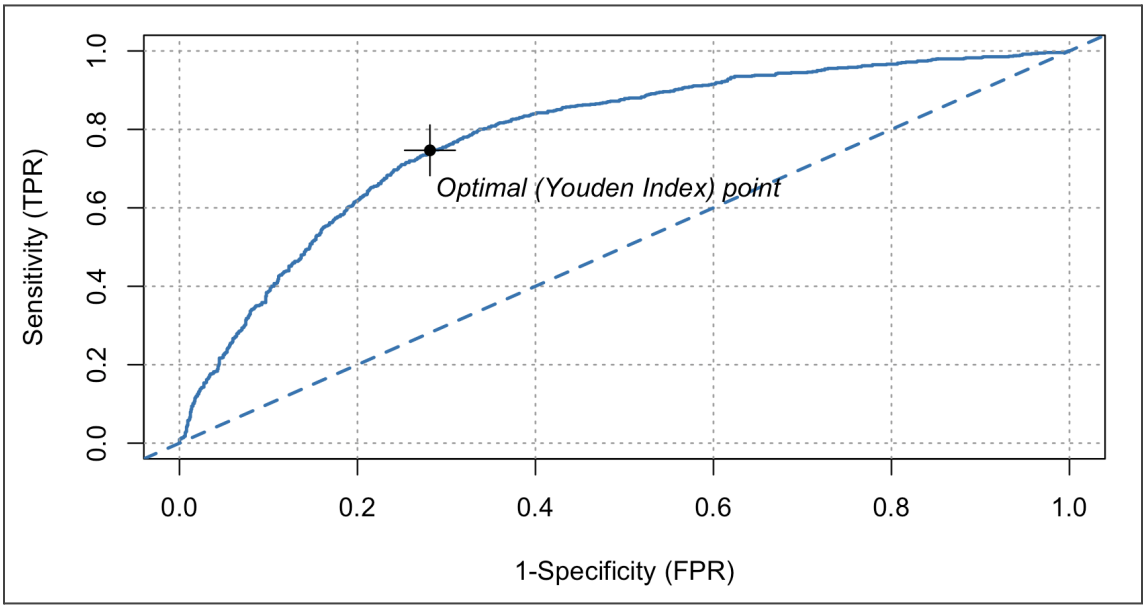

Figure A1: ROC Curve for Neural Network model